

PACK WITH LOVE CLUB

From Data to Impact

24-Lesson Project-Based Math Curriculum

Project-Based Math for Social Impact • 2026-2027 • Revised September 2026

Annual Challenge: Design, produce, market, sell, and evaluate a Pack with Love fundraising Squishy.

Learning Path: Mission → Business → Research → Product → Production → Sales → Analysis → Impact

Core Learning Cycle: Question → Prediction → Data → Math → Decision → Action → New Data → Reflection

Fieldwork (3): Survey Week (homework fieldwork, ≈ Oct 4-11) · Sales Event I · Sales Event II (January). All other sessions are Sunday classes.

Phase 1 · Understand the Business (3 sessions)

Lesson	Topic	Driving Question	Math & Thinking	Class Task	Deliverable
1	Understanding the Business I — Theory (Guest: Prof. Kevin Jialin Sun)	How does a product move from an idea to something a customer can actually receive?	Process, units, cost awareness, estimation	Professor explains the two pathways — U.S. domestic production vs. overseas production/import — and the full supply-chain framework. Students map the supply chain and mark cost, time, and risk points on each pathway.	Supply Chain Map + 2 Business Questions per student
2	Understanding the Business II — Theory Meets Practice (Guest: Tony Yao)	What does the textbook not tell us about real production and importing?	Order of magnitude, time, cost structure, risk comparison	Tony shares real order and supply-chain cases — MOQ, sampling, defects, delays. Students compare the theoretical process with real operations and answer questions collected in Lesson 1.	Reality Check Notes + Business Risk List
3	Understanding the Business III — Landed Cost in the Real World (Guest: Joyce Xie)	Why is the factory price never the final price?	Landed-cost intuition, percentages, currency awareness	Joyce walks through how procurement, production, quality, delivery, and local markets interact in real overseas operations. Students trace one product from factory gate to customer and list every cost it picks up on the way.	Landed-Cost Story Map (preview of Lesson 12)

Phase 2 · Market Research & Statistics (5 sessions incl. survey-week fieldwork)

Lesson	Topic	Driving Question	Math & Thinking	Class Task	Deliverable
4	Research Questions & a Survey That Can Be Trusted	If the product category is already Squishy, what decisions are still open — and how do we ask without pushing people toward our answer?	Statistical question, criteria, variables, categorical data, question design, bias	Define product-decision criteria (customer, age, shape, size, color, price, purchase intent); each student writes a hypothesis. Compare leading vs. neutral questions; teams draft an 8–10 question survey.	Prediction Sheet + Survey v1
5	Population, Sample & Bias — Sampling at School	If each of us asks 30–50 people at our own school, whom can the results represent?	Population, sample, random sample, stratified sample, sampling bias, fraction	Name random and stratified sampling. Build a school-based sampling plan: each student's school is a stratum; quota cards require mixed grades, genders, and non-friends. Simulate the bias created by asking only friends. Draft the School Support Request (students write the ask; an Adult Advisor submits it).	Sampling Plan + Quota Cards + School Support Request (draft)
6	Survey Launch Lab — fieldwork week	Will strangers answer the way we practiced?	Data collection, coding, tally, interviewer effect	In class: pilot the survey on classmates, practice neutral asking, code and tally pilot data, finalize quota cards. Homework (Survey Week, ≈ Oct 4–11): each student collects responses at their own school under the approved School Support Request, following school rules and the safety note.	Pilot Data + (due next week) Raw Data Set, 30–50 responses per team
7	From Raw Data to Percentages	If 21 people prefer “animal,” is that actually a lot?	Frequency, fraction, decimal, percentage	Combine every student's school data into one class dataset. Build a frequency table, convert counts to proportions/percentages, and check totals and errors. Compare: do different schools answer differently?	Clean Data Table + Percentage Summary
8	Graphs, Evidence & Product Recommendation	How can we make the market message understandable in 10 seconds?	Bar graph, pie chart, double bar graph, comparison	Create graphs, compare schools/grades/groups, and use Claim-Evidence-Reasoning to make a product recommendation. AI	Market Research Brief + Preliminary Product Decision

				checkpoint: let AI draft one chart, then critique what it missed or got wrong.	
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Phase 3 · Product, Production & Business Math (6 sessions)

Lesson	Topic	Driving Question	Math & Thinking	Class Task	Deliverable
9	Turn Data into a Product Specification	How does market data become a specification a factory can use?	Measurement, ratio, constraint	Decide shape, color, size, packaging, branding, and which requirements are non-negotiable versus flexible.	Product Specification Sheet
10	Cost per Unit	What does one product really cost?	Unit rate, fixed/variable cost, estimation	Break down sample, mold, packaging, production, shipping, and other costs; compare different MOQs.	Unit Cost Model
11	Compare Suppliers with Evidence	Is the cheapest supplier always the best supplier?	Weighted criteria, score, percentage weights	Build a weighted comparison using quality, price, lead time, MOQ, communication, and risk — with two live quotes on the table: a China supplier and a US wholesaler.	Supplier Comparison Matrix
12	Landed Cost & Import Math	Why is factory price not the same as final cost?	Rates, percent, landed cost, currency awareness	Combine production price, shipping, applicable duties/fees, and packaging into landed cost using current real tariff rates; compare with domestic wholesale. (Connects back to Lesson 3.)	Domestic vs. Import Cost Scenario
13	Pricing Scenarios	\$5, \$7, or \$10 — which price is actually better?	Revenue, gross margin, scenario analysis	Use willingness-to-pay survey data and unit cost to model different price/volume scenarios.	Pricing Recommendation
14	Production Decision & Quality Plan	Are we ready to spend real money and place an order?	Budget, quantity, buffer, percent defect	Set quantity, budget ceiling, quality checks, sample approval, and risk buffer; make a go/no-go recommendation. Adult Advisors place the actual order.	Production Approval Packet

Phase 4 · Marketing & Sales (6 sessions incl. 2 sales events)

Lesson	Topic	Driving Question	Math & Thinking	Class Task	Deliverable
15	Mission-Based Marketing	How can we sell the product	Message testing,	Write and test a value proposition that combines	Core Message + Sales Pitch

		without turning the mission into a slogan?	comparison	product value with the Pack with Love mission.	
16	Forecast Demand & Choose the Venue	How much should we bring — and where should we even set up the table?	Forecast, ratio, expected-value intuition, weighted criteria (reused from L11)	Use survey intent and expected traffic to build low/base/high forecasts per venue. Score candidate venues — school event, church gathering, farmers market — on traffic, audience fit, permission difficulty, and cost, using a weighted matrix. Students draft the outreach ask (email/call script); Adult Advisors make the actual contact and booking.	Sales Forecast + Venue Plan & Outreach Draft
17	Sales System & Event Day Kit	What must we prepare and record on sales day so the event runs — and teaches us something later?	Data fields, conversion rate, average revenue, checklists	Design the sales tracker, inventory log, and payment-record rules; assign booth roles. Build the Event Day Kit: booth checklist, tally sheet, pitch card, price card, inventory log, and customer feedback mini-cards. Confirm venue rules (school/market) and the cash-handling rule: an adult holds all money.	Sales Tracker + Team Roles + Event Day Kit
18	Sales Event I — fieldwork	Does real customer behavior match what people said in the survey?	Real-time tally, conversion, inventory	Run the first real sale using the Event Day Kit. Record inquiries, buyers, price, inventory change, and feedback.	Sales Data Set A
19	Analyze & Adjust Mid-Campaign	Should we keep the plan, lower price, change the pitch, or move locations?	Percentage change, comparison, prediction error	Compare forecast with actual results and choose one data-based adjustment for Event II.	Mid-Campaign Decision Memo
20	Sales Event II / Campaign Close — fieldwork	What happened after we changed the plan?	Cumulative total, rate, comparison	Run the second sale with the same recording system so the two events can be compared.	Final Sales Data Set

Phase 5 · Analyze, Reflect & Impact (4 sessions)

Lesson	Topic	Driving Question	Math & Thinking	Class Task	Deliverable
21	Post-Sale Statistical Analysis	What did we predict correctly, and where were	Difference, percent error, segment	Compare survey intent with actual purchase behavior across groups and events. AI	Post-Sale Data Dashboard

		we wrong?	analysis	checkpoint: AI may compute; every number it produces must be verified by hand before it enters the report.	
22	Profit, Net Fundraising & Impact Math	Why is sales revenue not the same as the amount we truly raised?	Revenue, cost, net proceeds, margin	Calculate revenue, direct product costs, approved expenses, and net fundraising; convert net fundraising into the backpack-impact metric on the shared goal.	Financial & Impact Summary
23	Business Review: What Would We Do Differently?	If we did this again, which decision would we change?	Evidence synthesis, trade-off	Review major decisions from survey through sales; distinguish a good decision with a poor outcome from a poor decision with a lucky outcome.	Business Review Report
24	Final Impact Showcase — March 7, 2027	Can we use evidence to explain the entire annual project clearly?	Data storytelling, visual communication	Present the mission, research, product, numbers, mistakes, improvements, and final impact to parents, community members, and supporters.	Final Presentation + Student Portfolio

Fieldwork Design & School Partnership

Survey Week (≈ Oct 4-11) — distributed fieldwork

- Each student surveys at their own school under an approved School Support Request — 15 students means samples from up to 15 different school communities, a stronger and more diverse sample than any single-site survey.
- Quota cards (from Lesson 5) require mixed grades, genders, and non-friends — the class's structural defense against friendship bias.
- **Safety & ethics note for parents:** surveys happen only at school among known peers or in supervised settings, following all school rules; no stranger outreach. Participation in the survey is voluntary and anonymous.

School Support Request — how the club asks a school

- Students draft the request in Lesson 5: who we are, what we are asking (a short voluntary survey during lunch/recess or advisory; later, permission for a small fundraising table at a school event), what we will and won't do (no names collected, no sales pressure, adult supervision).
- **An Adult Advisor reviews and submits every request** to the teacher, counselor, or administration — consistent with the Student Charter: students draft and recommend; adults make official commitments.
- Each approval is recorded (school, contact, date, scope) by the Project Recorder; fundraising at any school follows that school's own fundraising rules.

Sales Events I & II (January) — designed, not improvised

- **Venue Plan (Lesson 16):** candidate venues — a school event, a church gathering, a farmers market — scored on a weighted matrix (traffic, audience fit, permission difficulty, cost). Students draft the outreach email/call script; Adult Advisors make contact, handle any fees or forms, and book.
- **Event Day Kit (Lesson 17):** booth checklist · tally sheet · pitch card · price card · inventory log · customer feedback mini-cards — every student role has a sheet in hand; every transaction becomes a data point.
- Cash rule at every event: a supervising adult holds all money; students record, present, and sell. Net proceeds move through ACO's approved channel.
- Event dates are announced to families at least three weeks ahead; a student who cannot attend receives an alternative real role (pre-event prep or post-event analysis).

Program Parameters

- **Recommended grades:** Grades 6–7; elementary students welcome with an accompanying parent (support, not substitution). Content differentiates by mathematical maturity.
- **Core sessions:** 24 sessions — 21 Sunday classes (≈60 minutes) + 3 fieldwork activities (Survey Week homework fieldwork; Sales Events I & II). Sundays 5:00 PM Pacific / 8:00 PM Eastern.
- **Season:** September 13, 2026 – February 28, 2027; Final Impact Showcase March 7, 2027.
- **Tuition:** \$30 per session; \$720 for the 24-session season. Tuition pays for teaching and is not a donation.
- **Class size:** 15 students. Roles rotate through Market, Product, Finance, and Sales.
- **Participation:** Students do not need to travel to Africa. The full annual project is completed locally; the profit from class sales funds the Love & Hope Backpack Project's shared goal (net of real product costs).
- **Final assessment:** No traditional exam. Students complete a Student Portfolio, Business Review / Impact Report, and the Final Impact Showcase.
- **AI use:** AI may draft charts and compute — students must critique, verify, and take responsibility for every number. AI reduces unnecessary work, not necessary thinking.